Math Midterm · Practice Exam 2 (Challenge)

Harder questions · strict on signs, exact forms and complete answers

Name: Date: Period:

Time: 65 minutes · 80 points. Part A: 18 multiple-choice questions in three levels, 2 points each (36 points). Circle the letter of the best answer. Part B: 5 free-response questions (44 points). Show all your work in the space given: a correct answer with no work earns at most 1 point. "Exact" means leave answers as fractions and simplified radicals. A scientific or graphing calculator may be used; no formula sheet is provided.

Standard normal table (area to the LEFT of z). Use these values where a question says "use the table".

z	−2.00	−1.50	−1.00	−0.50	0.50	1.00	1.28	1.29	1.50	1.75	2.00
area	0.0228	0.0668	0.1587	0.3085	0.6915	0.8413	0.8997	0.9015	0.9332	0.9599	0.9772

Part A · Multiple Choice (2 points each, 36 points)

LEVEL 1 · RECALL AND CORE SKILLS

1. What is the value of -2^4?

A 16 **B** -16

C -8 **D** $\frac{1}{16}$

2. Simplify $\left(\frac{2}{3}\right)^{-2}$.

A $\frac{4}{9}$ **B** $-\frac{4}{9}$

C $-\frac{9}{4}$ **D** $\frac{9}{4}$

3. Which is the complete factorization of $x^3 - 27$?

A $(x-3)(x^2+3x+9)$ **B** $(x-3)(x^2-3x+9)$

C $(x-3)^3$ **D** $(x+3)(x^2-3x+9)$

4. If $a+b=7$ and $ab=10$, what is a^2+b^2?

A 49 **B** 39

C 29 **D** 69

5. Which of these is a parameter?

A The mean score of 50 tests chosen at random from a district's 2,000 tests

B The percent of all 1,200 students at a school who play a sport, taken from school records

C The percent of 300 surveyed voters who support a school bond

D The median height of a random sample of 25 trees in a park

6. Researchers randomly assign 40 volunteers to take a vitamin and 40 to take a placebo, then compare how many colds each group gets. This is

A an experiment
B a survey
C an observational study
D a simulation

LEVEL 2 · APPLY

7. Solve $9^{x} = 27^{x-1}$.

A $x = 1$
B $x = -3$
C $x = \dfrac{3}{5}$
D $x = 3$

8. $(x - 2)$ is a factor of $2x^3 + kx^2 - 5x + 2$. What is k?

A 1
B -2
C 2
D -4

9. What value of c makes $x^2 - 12x + c$ a perfect square trinomial?

A 6
B 144
C 36
D -36

10. $kx^2 + 6x + 1 = 0$ has two equal real roots. What is k?

A 9
B 36
C -9
D 6

11. A parabola has x-intercepts at -1 and 7. What is its axis of symmetry?

A $x = 4$
B $x = 3$
C $x = -3$
D $x = 6$

12. SAT scores are normal with $\mu = 500$ and $\sigma = 100$. About what percent of students score above 700?

A 5%
B 47.5%
C 97.5%
D 2.5%

LEVEL 3 · MULTI-STEP AND REASONING

13. r and s are the roots of $x^2 - 5x + 3 = 0$. What is $r^2 + s^2$?

A 19
B 25
C 31
D 22

14. Solve $4^{x} - 10 \cdot 2^{x} + 16 = 0$.

A $x = 2$ or $x = 8$
B $x = 1$ or $x = 3$
C $x = 1$ or $x = 8$
D $x = -1$ or $x = -3$

15. Which statement about $y = 2(x + 3)^2 - 5$ is true?

A Vertex $(3, -5)$; it has a minimum.
B Vertex $(-3, -5)$; maximum value -5.
C Vertex $(-3, 5)$; it has a minimum.
D Vertex $(-3, -5)$; minimum value -5.

16. A ball's height is $h = -16t^2 + 64t + 5$ (feet, t in seconds). What is its maximum height?

A 64 ft
B 197 ft
C 69 ft
D 5 ft

17. A pollster wants a margin of error of ±2%. About how many people should be in the random sample?

A 50
B 250
C $10{,}000$
D $2{,}500$

18. Scores are normal with $\mu = 75$ and $\sigma = 8$. A score at the 90th percentile has $z \approx 1.28$. About what is that score?

A 65
B 83
C 85
D 88

Part B · Free Response (44 points)

19. Find the mistakes (8 points)

Each solution below has a mistake. Describe the mistake in words, then write the correct answer.

(A) Simplify $(3x^{-2})^{-2}$. **(2 points)**

STUDENT'S WORK
$(3x^{-2})^{-2} = 3x^{4}$

Mistake and correct answer:

(B) Factor $6x^2 - x - 12$. **(2 points)**

STUDENT'S WORK
$ac = -72$. Two numbers with product -72 and sum -1: -9 and 8.
$6x^2 - 9x + 8x - 12 = 3x(2x - 3) + 4(2x - 3) = (3x - 4)(2x - 3)$

Mistake and correct answer:

(C) Solve $(x - 3)(x + 2) = 6$. **(2 points)**

STUDENT'S WORK
$x - 3 = 6$ or $x + 2 = 6$, so $x = 9$ or $x = 4$.

Mistake and correct answer:

(D) Test scores are normal with $\mu = 60$ and $\sigma = 10$. What percent scored above 75? **(2 points)**

STUDENT'S WORK

$z = \frac{75-60}{10} = 1.5$. The table gives 0.9332, so 93.32% scored above 75.

Mistake and correct answer:

20. Quadratic equations in depth (10 points)

(a) Solve $2x^2 + 3x - 7 = 0$ by completing the square. Give exact answers. **(3 points)**

Work:

(b) Find <u>both</u> values of k for which $x^2 + kx + 9 = 0$ has exactly one (repeated) real root. **(2 points)**

Work:

(c) One root of $x^2 - 7x + c = 0$ is 3. Use the sum and the product of the roots to find the other root and the value of c. **(2 points)**

Work:

(d) Solve $9^x - 4 \cdot 3^x + 3 = 0$. **(3 points)**

Work:

21. Building and using quadratic models (10 points)

A parabola has vertex $(2, -8)$ and passes through the point $(0, -4)$.

(a) Write its equation in vertex form. **(2 points)**

Work:

(b) Write the equation in standard form. **(1 point)**

Work:

(c) Find the x-intercepts. Give exact answers, then decimals to two places. **(2 points)**

Work:

A model rocket is launched straight up from the ground at 80 ft/s. Its height after t seconds is $h(t) = -16t^2 + 80t$ feet.

(d) At what times is the rocket 64 ft above the ground? Explain why there are two answers. **(3 points)**

Work:

(e) Find the maximum height, and the time when the rocket lands. **(2 points)**

Work:

22. Grouped data (8 points)

The table shows how many minutes 40 students spent on homework one night. (A class such as 10–20 means 10 or more but less than 20.)

Minutes	0–10	10–20	20–30	30–40	40–50
Number of students	5	8	12	9	6

(a) Find the mean, using the class marks (midpoints). **(3 points)**

Work:

(b) Find the median. **(2 points)**

Work:

(c) Find the mode. **(2 points)**

Work:

(d) Compare your mean, median and mode. Is the distribution roughly symmetric or clearly skewed? Explain. **(1 point)**

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23. The normal distribution and inference (8 points)

The heights of 9th-grade boys at a large school are normally distributed with $\mu = 66$ inches and $\sigma = 3$ inches.

(a) Use the empirical rule to find the percent of boys taller than 69 inches. **(1 point)**

Work:

(b) Use the table to find the percent of boys shorter than 61.5 inches. **(2 points)**

Work:

(c) The tallest 10% are invited to a basketball tryout. Use the table to find the minimum height for an invitation, to the nearest tenth of an inch. **(2 points)**

Work:

(d) A company claims that 50% of teens prefer its drink. In a random sample of 100 teens, 38 prefer it. Find the margin of error and the interval for the true percent. In 200 simulations of samples of 100 teens with a true rate of 50%, a result of 38 or fewer came up 3 times. At a 5% significance level, should the company’s claim be rejected? Explain using both results. **(3 points)**

Work: